

MUSTAFA SHOUKAT

Senior AI Engineer · Enterprise Architecture & Technical Leadership · Agentic RAG

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SUMMARY

AI Engineer operating at Staff-level scope: owns architecture end to end for distributed, fault-tolerant AI platforms, not just individual features. Architected and shipped 20+ production systems for enterprise and government-grade clients across Saudi Arabia, the UAE, the US, and the UK, including a multi-tenant, distributed RAG platform with 5-level RBAC, full audit logging, and on-premises/air-gapped deployment for regulated networks, and a Claude-as-judge fine-tuning pipeline that took an Arabic LLM to #1 on the AraGen Leaderboard. 5+ years across Python, Java, FastAPI, scalable platform APIs, vector databases, SQL/NoSQL data stores, cloud-native deployment (AWS, GCP, Azure, Docker), and production observability (Prometheus/Grafana/Loki), building agentic systems (LangGraph, CrewAI, MCP) with tool-calling and multi-step planning. Set engineering standards, trained 150+ professionals, and mentored engineers in AI engineering; active open-source contributor to LangChain, RAGAS, and Unstructured.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, Rust

AI / LLM & Orchestration: LangChain, LangGraph, CrewAI, LlamaIndex, AutoGen, MCP (Model Context Protocol), tool-calling & multi-step planning, OpenAI, Gemini, Groq, Cohere, HuggingFace, Ollama

RAG & Vectors: Qdrant, Milvus, FAISS, ChromaDB, Pinecone, pgvector; hybrid/semantic retrieval (BM25 + dense), RRF, rerankers, NLI faithfulness scoring

Evaluation / Governance: LangSmith, Helicone, MLflow; RAG evaluation & groundedness scoring, guardrails, prompt-injection defense, Claude-as-judge pipelines

Backend: FastAPI, Django, Flask, Node.js/Express, Celery, Redis, WebSockets, SSE

Cloud / DevOps: Docker, Docker Compose, GitHub Actions CI/CD, AWS (S3, App Runner), GCP, Azure, Nginx, Prometheus, Grafana, Loki, Sentry

Databases: PostgreSQL, MongoDB (SQL & NoSQL data stores), Redis, Supabase, Neon; SQLAlchemy, Drizzle, Prisma, Alembic

Specialties: Multi-tenant, distributed, fault-tolerant architecture (Celery/Redis), scalable platform APIs, air-gapped RAG, RBAC/access control, Arabic-first NLP

EXPERIENCE

AI Software Engineer, Independent / Contract · Remote (KSA / UAE / UK / US) | 2024 to Present

- Ship and maintain production AI systems across RAG platforms, voice AI, WhatsApp automation, and CRMs for clients in four countries: deployed systems, not prototypes.
- Built a government-contracting platform that scrapes SAM.gov by NAICS code and runs LangGraph agents for solicitation analysis, bid/no-bid decisions, and automated PDF proposal generation on a provider-agnostic LLM layer.
- Engineered fault-tolerant WhatsApp lead-qualification systems with fail-closed HMAC webhook verification, Whisper voice-note transcription, atomic message deduplication, and CRM dashboards with human handover.
- Advise organizations moving AI from pilot to day-to-day operations: architecture, evaluation, scalability, and cost-optimized inference.

Agentic AI Engineer, Watad (Navid) · Riyadh, Saudi Arabia | 2023 to 2026

- Owned end-to-end architecture** for ByanRAG 2.2, a production, multi-tenant Arabic-first RAG platform, with org-isolated Qdrant collections, a 5-level Casbin RBAC hierarchy enforced on every endpoint, and a 14-panel admin console (users, documents, analytics, audit logs, system health), built for on-premises deployment inside regulated networks.
- Designed the hybrid semantic retrieval pipeline** (dense embeddings, BM25 sparse search, RRF fusion, CrossEncoder reranking) powering source-cited, bilingual (Arabic/English) streaming SSE chat.
- Designed a Claude-as-judge reward pipeline** (3C3H metric) and NLI-based evaluation guardrails, including faithfulness/groundedness scoring, Recall@5, MRR, and prompt-injection defenses, that took Yehia R1 to **#1 on the AraGen Leaderboard** (0.5B–25B).
- Led multi-agent orchestration** (LangGraph, CrewAI) with tool-calling and multi-step planning to automate document analysis, bid/no-bid decisions, and proposal generation for GCC enterprise clients, replacing manual analyst workflows.
- Owned the production stack end to end** and set engineering standards for testing, deployment, and model integration across the team: Docker Compose, nginx/TLS, LangSmith tracing, full Prometheus/Grafana/Loki observability, and a 600+ test suite with CI/CD.

Generative AI Consultant, Aretec, Inc. · Remote (US) | Jul 2023 to Aug 2024

- Built a production Text-to-SQL RAG application enabling non-technical business users to query enterprise databases in plain English, combining SQL agents, LangChain retrieval, and PEFT-tuned LLM fine-tuning for production-level accuracy and response speed.
- Engineered query-understanding and retrieval design to handle ambiguous natural-language queries against complex schemas, deployed for internal business intelligence.

Data Scientist, COMET Estimating LLC · Austin, TX (Remote) | Jul 2021 to Jun 2023

- Built ML, NLP, and automation pipelines for a construction-industry data company: predictive analytics models, document-processing pipelines, and intelligent estimation tools.
- Hardened pipelines against messy real-world data: inconsistent formats, missing fields, and high-variance inputs.

SELECTED ARCHITECTURE WORK

- ByanRAG 2.2 · Multi-Tenant Arabic Document Management & RAG Platform** (FastAPI, React, TypeScript, Qdrant, Casbin, Docker, Prometheus/Grafana/Loki): Enterprise document management system with grounded, source-cited Arabic-first Q&A. Multi-tenant by design with per-organization data isolation, a 5-level Casbin RBAC hierarchy, and a 14-panel admin console; architected for on-premises deployment inside regulated and government networks. Hybrid retrieval combining dense embeddings, BM25 sparse search, RRF fusion, and CrossEncoder reranking, with NLI faithfulness scoring. Containerized with Docker Compose, deployed behind nginx with TLS and full observability, backed by a 600+ test suite and CI/CD.
- Global Holman · AI Government-Contracting Platform** (FastAPI, LangGraph, LlamaIndex, Next.js, Celery/Redis, async Postgres): End-to-end agentic platform that scrapes SAM.gov by NAICS code, ingests solicitation PDFs into a RAG pipeline, and runs LangGraph multi-agent workflows for requirement analysis, bid/no-bid decisions, and automated PDF proposal generation, over a provider-agnostic LLM layer, fault-tolerant Celery/Redis task processing, reusable platform APIs, and swappable vector stores.
- Navid RAG ARC · Arabic LLM RAG Service** (FastAPI, LangChain, Milvus, Celery, Arabic NLP, sentence-transformers): Modular Arabic-first RAG backend with hybrid BM25 + dense retrieval, Reciprocal Rank Fusion, multilingual reranking, and a tool-calling ReAct agent for multi-step reasoning and planning.

MENTORSHIP & OPEN SOURCE

- Trained **150+ professionals** and mentored engineers in Data Science, Machine Learning, AI Engineering, and system architecture across enterprise and independent settings.
- LangChain** (92k+ stars): fixed malformed function-call parsing + added tests (merged PR).
- RAGAS** (7k+ stars): multiple merged PRs, including bug fixes, multilingual (Arabic/CJK) support, and backward-compatible import aliases, plus tests.
- Unstructured** (14k+ stars): PaddleOCR language-mapping fix + tests (merged PR).

EDUCATION

MS Artificial Intelligence · National University of Sciences and Technology (NUST), Pakistan, In Progress (2025–Present)

BS Computer Science · Virtual University of Pakistan, Completed

Kaggle active contributor | Hugging Face model cards | AI / Data Science certifications | Former Student Representative